



Hydrogen peroxide decomposition in serpentine mini channel with silver catalyst

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ABSTRACT

Micro thrusters were used to produce thrust in the order of a few Newtons to position satellites and other designed space objects. For such applications, monopropellant thrusters using hydrogen peroxide are preferred for their high maneuverability, accurate positioning, and attitude control. Hydrogen peroxide is a green monopropellant that forms the products (water and vapor) after it gets decomposed in contact with a catalyst. Complete catalytic decomposition of hydrogen peroxide is still a challenging task. In the present work, 30% of hydrogen peroxide concentration with silver catalyst (modified geometry) were used to increase reaction rate. The effect of silver placed at different locations in serpentine and straight mini channels (same dimensions) is analyzed. The analysis is done for a varied number of silver catalysts placed at different locations in serpentine and straight mini channels having a diameter of 0.25 cm. The serpentine mini channel shows better decomposition of hydrogen peroxide especially at shorter length of 10 cm than the straight mini channel with the same length of catalyst because of its better mixing capability due to non-uniformity in the flow field, especially at the bend sections. The results will be useful in determining the appropriate geometry and placement of the catalyst for efficient monopropellant decomposition.

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1. Introduction

Micropropulsion systems are being developed in space propulsion, specifically for satellite systems that require controlled thrust and low specific impulse. Such systems are used for precise adjustments, attitude and orbital control of satellites. This technology has several advantages that includes lower operational cost, long life, occupying less space and the possibility to replace the traditional larger satellites with a number of mini or micro satellites with minimum fuel consumption. This technology overcomes the challenges that are faced by the conventional propulsion systems that employ combustion. But the most challenging part of microscale propulsion is achieving a precise thrust in the order of a few Newtons. Green monopropellant thrusters like hydrogen peroxide, which are less toxic and decompose in the presence of a catalyst are highly preferred instead of other conventional toxic propellants. Among the various catalysts, silver is the most commonly used material. The standardized Micro /Mini channel design with

appropriate catalyst arrangement to produce a required level of thrust does not exist at present. The purpose of this paper is to design a mini channel with various catalyst arrangements which can give complete decomposition of 30% hydrogen peroxide. The initial research and tests were conducted during World War II at Germany on hydrogen peroxide propulsion for production of thrust [1]. Many flight tests were performed for production of different level of thrust based on their range, performance and other factors in aircrafts, missiles and rockets. Decomposition of hydrogen peroxide using silver-gold alloys as catalyst for various concentrations were investigated [2].

Based on the catalyst oxygen affinity, gold and silver have largely similar properties. Therefore, silver was selected as the catalyst material in the study. The decomposition of H_2O_2 using various catalysts combinations in two type of reactors (constant volume-pressure) was studied and their performance was detailed [3]. A computational model was developed to improve the MEMS (Micro Electro-Mechanical system) based micro thruster that uses hydrogen peroxide as propellant and decomposed by catalyst to generate thrust and specific impulse [4]. The study results show that decomposition of H_2O_2 occurs slowly due to non-diffusive

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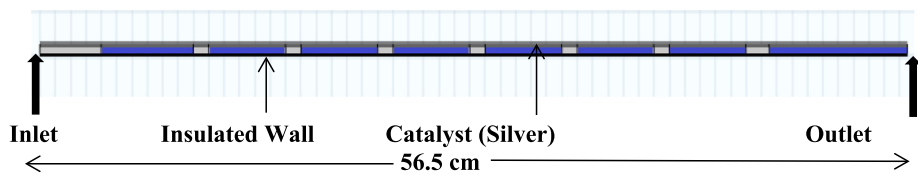


Fig. 1. Straight Mini channel.

mechanism. The reaction rate was increased by introducing gas bubbles into the flow and that enhanced the decomposition due to its mixing characteristics. The small satellites systems were

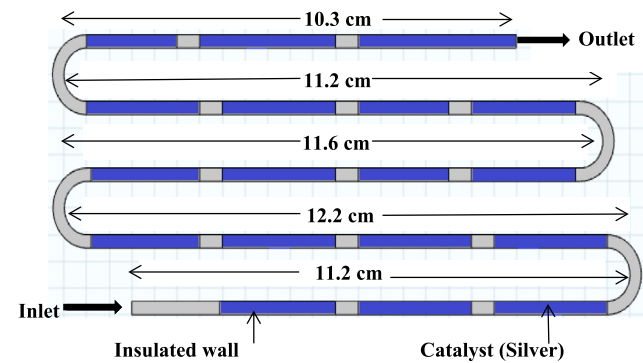


Fig. 2. Serpentine Mini channel.

Table 1
Concentration value of 30% of Hydrogen Peroxide (H_2O_2).

Composition	Initial Concentrations (mole/ m^3)
30% of Hydrogen Peroxide (H_2O_2)	8823
Water (H_2O)	38,888
Oxygen (O_2)	0

assessed in which also included economic categories as efficiency and return [5].

A numerical study for the decomposition of hydrogen peroxide on the helical shaped silver catalyst was detailed in [6]. It was found that increasing the number and shapes of the catalyst would give better decomposition. It was observed that the reaction rate can be increased through proper catalytic design [7]. The flow rate played a major role in terms of decomposition to produce precise level of thrust. The product species concentration was measured using a 3D straight channel model to examine the impact of the number and placement of catalysts (Annular silver catalyst) on the decomposition of 30% hydrogen peroxide [8]. Monopropellant trade studies for hydrazine and hydrogen peroxide were reported in [9]. Mezyk et al. [10] designed a decomposition process to get 98% hydrogen peroxide which was utilized for low thrust propulsion systems. A detailed kinetic study for decomposition of hydrogen peroxide using bimetallic catalysts was done in [11]. Reid et al. [12] used additive manufacturing technique to print a ceramic catalyst for hydrogen peroxide decomposition recently. Green monopropellant hydrogen peroxide was recommended by researchers to replace the highly toxic propellant like hydrazine to maintain the eco-friendly environment. The present work discuss about the effect of serpentine and straight mini channel for the better hydrogen peroxide decomposition with silver catalyst placed at different locations. The effect of length of catalysts ranging from 10 cm to 45 cm placed at different locations on catalytic decomposition is studied. Serpentine mini channels are proposed for better mixing

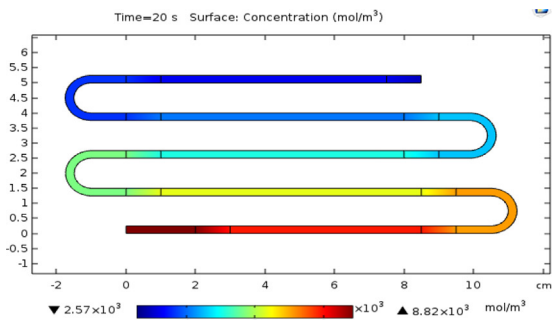


Fig. 3. Surface concentration variation of H_2O_2 with 10 cm catalyst (mole/ m^3).

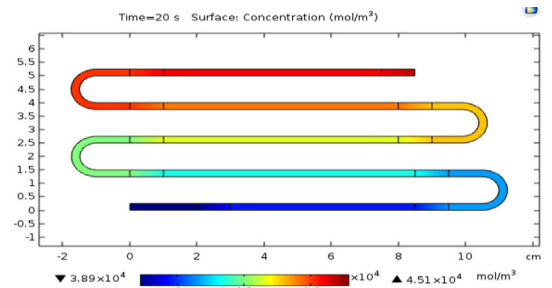
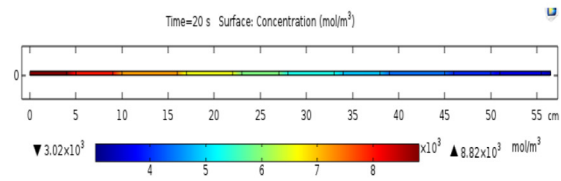


Fig. 4. Surface concentration variation of H_2O with 10 cm catalyst (mole/ m^3).

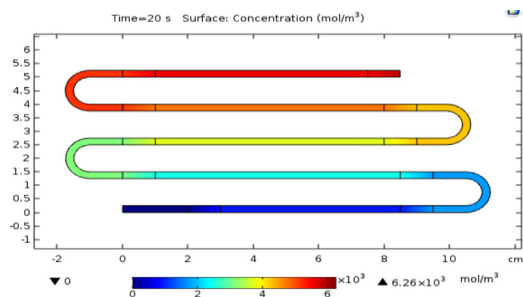
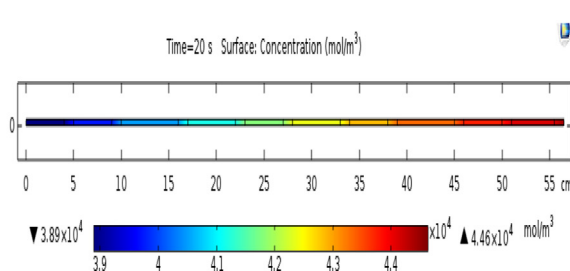


Fig. 5. Surface concentration variation of O_2 with 10 cm catalyst (mole/m^3).

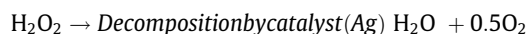


enhancement that increased the decomposition when compared to straight mini channels.

2. Methodology

The model is developed as a validation exercise to justify the hydrogen peroxide decomposition characteristics. For the 0-D

model, the reaction is assumed to take place in a batch-type reactor (constant volume). The Arrhenius expression and first-order rate expression are used for the concentration reaction [6]. A locked concentration is retained for the surface species (catalyst). The serpentine mini channel (unfolded) and straight mini channel with the same dimensions of [0.25 cm in diameter & 56.5 cm in length] are used. The materials used are borosilicate glass (wall), hydrogen peroxide (liquid) and silver (catalyst) and their respective properties are used in the model. An annular-shaped silver catalyst is placed inside the serpentine and straight mini channels at different locations with varied dimensions. The decomposition reaction is



In the next step, the 2D models of serpentine and straight channels are analyzed using COMSOL Multiphysics with a diameter of 0.25 cm and a length of 56.5 cm. The analysis is done for an inlet velocity of 0.1 m/s for 30% hydrogen peroxide concentration. Silver catalysts of length of 10, 20, 30, 40, and 45 cm placed at different locations in serpentine and straight mini channels are arranged.

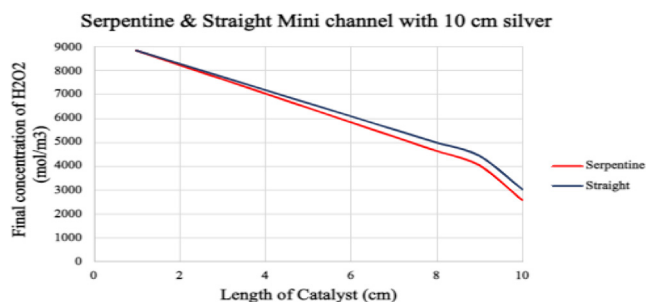


Fig. 6. Final Concentration of H_2O_2 for 10 cm silver.

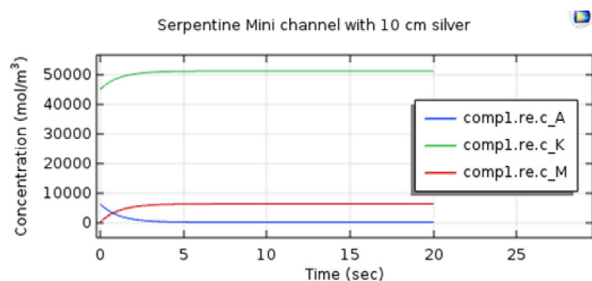


Fig. 7. Time vs Concentration decomposition plot.

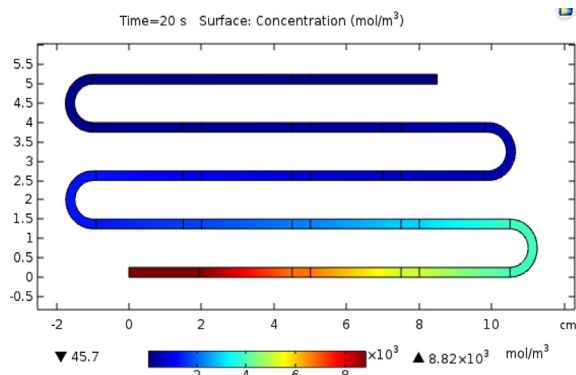
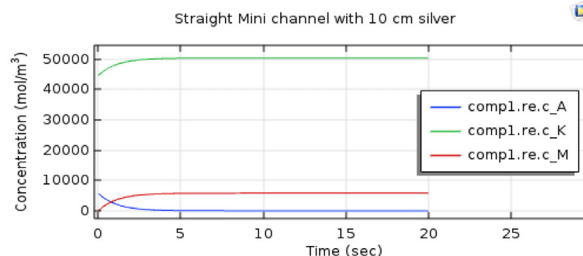


Fig. 8. Concentration variation of H_2O_2 with 10 cm catalyst (mole/m^3).

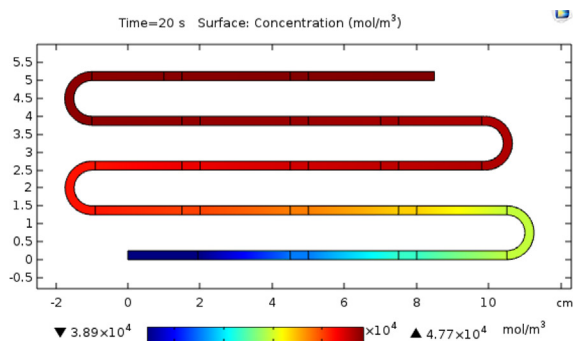


Fig. 9. Concentration variation of H₂O with 45 cm catalyst (mole/m³).

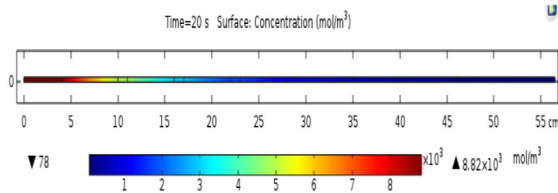


Fig. 10. Concentration variation of O₂ with 45 cm catalyst (mole/m³).

The flow is set to be laminar flow and the respective concentration inflow value is given for 30% hydrogen peroxide. Time-dependent studies are done for 0 to 20 s, with a time step of 0.1 s. The serpentine and straight mini channel dimensions and their boundary conditions are mentioned in Figs. 1-2. A detailed validation study is done for flow through a pipe of same dimensions in which the velocity profiles and pressure drop is compared and matched with the analytical model with less than 1% difference for a grid independent fine mesh. The obtained results are also compared and validated with similar works of [6] and [8] done for straight mini channels.

2.1. Governing equations

Arrhenius equation is used to measure the surface concentration species (H₂O₂, H₂O and O₂) after they get decomposed with silver catalyst(1)

$$K = A \cdot (T/T_{ref})^n \cdot e^{(-E/RT)} T_{ref} \quad (1)$$

Where A is the Reaction's frequency factor (0.18×10^9 m³/s mol);

T is the given Temperature;

T_{ref} is the Reference temperature;

n is the forward temperature exponent;

E is the reaction's activation energy (52200 J/mol);

R is the gas constant (8.314 J/mol);

K is the Reaction rate.

Consideration is given to the convection transfer mechanism. The reported species diffusion coefficients are 4×10^4 m/s. For the components under consideration, the given data and values are constants. Table 1 shows the concentration value of 30% hydrogen peroxide which is given as an inflow concentration values in transported of diluted species to study the decomposition.

The boundary conditions are given as follows: the inlet velocity being specified as 0.1 m/s and pressure is given as 1 atm at the outlet. The flux discontinuity condition is specified over the catalyst boundary for the surface reaction between hydrogen peroxide and silver to happen [8].

3. Results and discussion

The computational analysis is performed to study the decomposition of hydrogen peroxide in serpentine and straight mini channels with varied length of silver catalyst at different locations. The following sections present the results in terms of concentration for various catalyst lengths.

The concentration variations of H₂O₂, H₂O, and O₂ for a 30% concentration of hydrogen peroxide when passing through the 10 cm silver catalyst in serpentine and straight mini channels at different locations for 20 s are shown in Figs. 3-5. The concentration plot shows that the concentration of H₂O₂ decreased 70% in serpentine

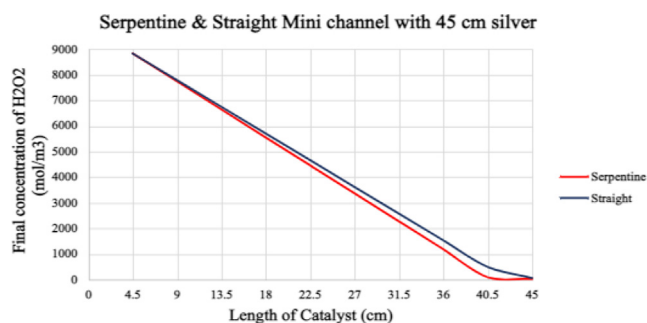


Fig. 11. . Final Concentration of H₂O₂ with 45 cm silver.

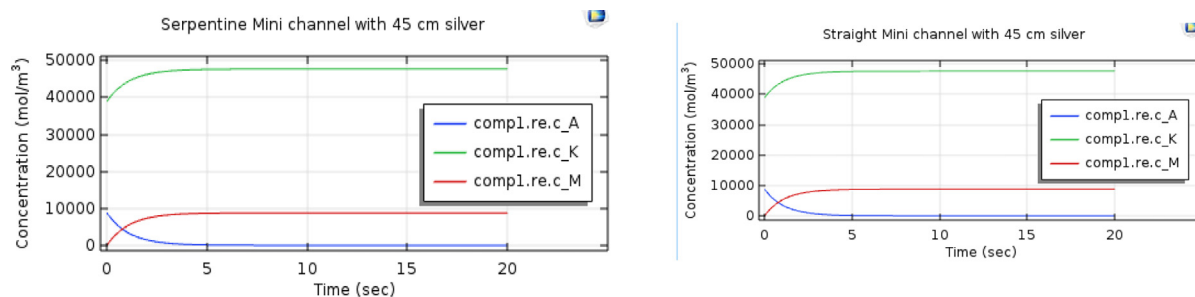


Fig. 12. Time vs concentration decomposition plot.

Table 2

Final concentration variation based on the length of catalysts.

Sl. No	Length of Catalyst (cm)	Serpentine final concentration in (mol/m ³)			Straight final concentration in (mol/m ³)			Decomposition Rate of H ₂ O ₂ (%)	
		H ₂ O ₂	H ₂ O ($\times 10^4$)	O ₂	H ₂ O ₂	H ₂ O ($\times 10^4$)	O ₂	Serpentine Mini channel	Straight Mini channel
1.	10	2570	4.51	6260	3020	4.46	5810	70	65
2.	20	834	4.69	7990	1050	4.67	7810	91	88
3.	30	264	4.74	8560	384	4.73	8450	97	95
4.	40	77.5	4.76	8750	140	4.75	8680	99	98
5.	45	45.7	4.77	8780	78	4.76	8740	99.50	99

and 65.7% in straight mini-channel while the H₂O and O₂ concentration increased consequently. Decomposition rate of hydrogen peroxide in serpentine is 6.14% more than in straight channel for the same length of catalyst due to its improved mixing capability, particularly at the bend section. The maximum velocity of 0.14 m/s is observed at the bend sections. Radial pressure gradient arises in the flow field, whenever the fluid flows through bend sections due to centrifugal force acting on it. The fluid which flows at the pipe center will move towards the inner side of the wall and turning around the outer side due to the pressure gradient present in it. The adverse pressure gradient at the inner wall and immediately downstream of the bend may produce the flow separation when the bend curvature ratio is very small [$R_c/D = 0.5$], which can highly increase pressure losses. [13]. In serpentine channel, placing the catalyst in the U-bend section does not give effective results because the flow is moving faster at that section and that makes less contact with the catalyst. It is also observed that the hydrogen peroxide is not completely decomposed inside the mini channel. (See Fig. 6).

The reaction-engineering graph in Fig. 7 shows that the hydrogen peroxide concentration get decreases over the time and reaches a zero value. It decomposes the 6253 mol/m³ in serpentine and 5803 mol/m³ in straight mini channel' is observed and the corresponding concentrations of water and oxygen increases. The pat-

terns of all the concentration species gets change after 1 s, is shown in Fig. 7. After 5 s, the concentration of each product started to stabilize and 10 cm of catalyst was not enough to achieve full decomposition. So the length of the catalyst needs to be increased to get complete decomposition inside the mini channel. Similarly, the concentration variations for serpentine and straight mini channel with 45 cm of catalyst are shown in Figs. 8–10. (See Figs. 11 and 12).

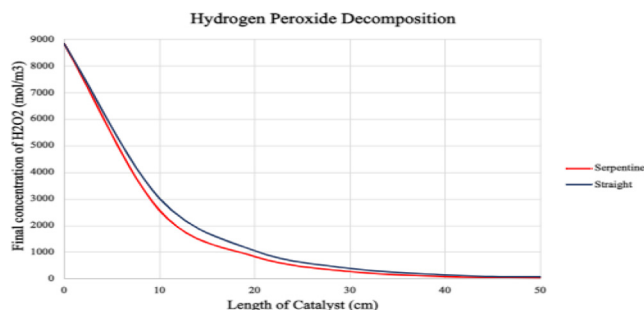
The decomposition rate of H₂O₂ in serpentine is 0.50% more than in straight channels for a 45 cm length of catalyst. The analysis was done for an increased length of catalyst from 10 cm to 45 cm in order to enhance the hydrogen peroxide decomposition. The results of final surface concentrations for various lengths of catalyst are summarized in Table 2 and plotted in Fig. 13.

4. Conclusion

A numerical model using COMSOL Multiphysics 5.3a is developed to study the effects of geometry, number, length and catalyst arrangement for the 30% of hydrogen peroxide decomposition. The serpentine mini channel shows better decomposition characteristics than the straight channel for the same length of catalyst. The catalyst is placed at different locations instead of one location to increase the rate of reaction. In serpentine, placing the catalyst in the U-bend section does not give effective results because the flow is moving faster at that section and that makes less contact with the catalyst. The geometry model with a 45 cm silver catalyst in a serpentine mini channel shows 99.50% decomposition due to its enhanced mixing capability. Increasing the length of the catalyst from 45 to 50 cm does not give complete decomposition due to less reaction effectiveness. Furthermore, the future work will focus on the varying different parameters like geometry modifications of mini channels, catalyst (size, shape, and arrangement), varying concentrations, and flow rate and inlet velocity to obtain complete decomposition of hydrogen peroxide.

Data availability

Data will be made available on request.

Fig. 13. Final Concentration of H₂O₂ vs Length of catalyst.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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